



ME 300: Engineering Thermodynamics

Example: Energy Conservation in Ideal Gases

A closed piston-cylinder system filled with air rejects 25 kJ of energy in a heat-interaction while experiencing a volume change of 0.1 m^3 (from 0.15 to 0.05 m^3). Assuming a reversible, constant-pressure process at 350 kPa , determine the change in energy of the air inside.

Given : $M = \text{const.}$

$${}_1Q_2 = -25000 \text{ J}$$

$$V_1 = 0.15 \text{ m}^3$$

$$V_2 = 0.05 \text{ m}^3$$

$$P_2 = P_1 = 350000 \text{ Pa}$$

Find : $\Delta E = \Delta \bar{U}$

Assume : ideal gas

: quasi-equilibrium

$$: \Delta KE = \Delta PE = 0$$

$$\Rightarrow \Delta \bar{U} = {}_1Q_2 - {}_1\bar{W}_2$$
$$= = =$$



Which of the following assumptions is *NOT* appropriate for this problem?

- A. Quasi-equilibrium
- B. $\Delta KE = \Delta PE = 0$
- C. Ideal gas
- D. They are all appropriate

Energy Conservation

$$\Delta U = {}_1Q_2 - {}_1\bar{W}_2 \rightarrow {}_1\bar{W}_2 = \int_{v_1}^{v_2} P dv = P \int_{v_1}^{v_2} dv = P(v_2 - v_1)$$
$$= 350000(0.05 - 0.15)$$
$$= -35000$$

$$\Delta U = -25000 - (-35000)$$
$$= 10000 \text{ J}$$